Searching for Correlates in fMRI data

Motivation:

The purpose of this assignment is to explore how a data analysis workflow can influence the validity of studies, and show how you can use simulations to validate an analysis workflow before you even apply it to real data, or even before you collect data. We will use an example from cognitive neuroscience. There are many studies in which researchers search for brain regions whose activity (in some task) is correlated with a behavioral or psychological measure. Many such studies use the following general workflow: first identify the brain regions whose activity are correlated with the psychological or behavioral measure, and then average the activity in these identified regions to measure how well this aggregate brain activity score correlates with the cognitive trait. You will be simulating this type of experiment and analysis, in a case where ground truth is no effect, vs. a case where ground truth is a real effect.

Preparations:

Suppose you tested N=12 subjects in a study involving a psychological test and a brain scan.

First, for each subject you perform a psychological test for some personality trait. Let's call their test score T. To simulate this, assign a score to each subject, drawn at random from a normal distribution with a with a mean of μ_T and a standard deviation of σ_T. [Try mean=50, std=10]

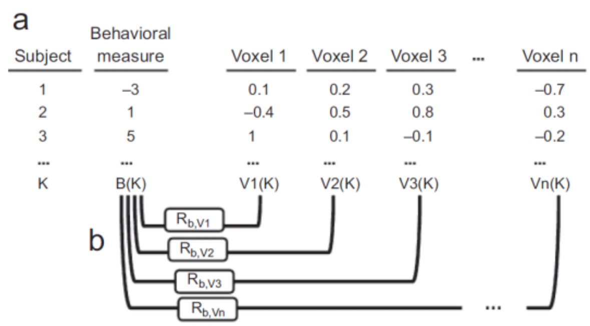
Then for each subject simulate an fMRI brain scan, in which we imagine that you measured the increase in brain activity (BOLD signal) in some test condition, relative to a control condition ("differential activity"). The differential activity is measured separately in each of N_{vox} voxels (cubic volumes in the brain). For simplicity, assume that you are able to find the corresponding voxels in all N=12 brains, and the same voxels are modulated by the task in all the subjects. [Try N_{vox} =5,000].

Simulate that 10% of the voxels really are affected by the task condition in the scanner ("task dependent" voxels). You can simulate their values by drawing from a normal distribution with $\mu_A,\ \sigma_A$. Assume the rest of the voxels have no difference in activity ("task independent" voxels). Simulate their values by drawing from a normal distribution with $\mu_B = 0, \sigma_B = \sigma_A$. [try $\mu_A = 1,\ \sigma_A = 1$]

In the end you'll have N_{vox} values (voxels) for each of the N=12 subjects. Here's the point: although the "differential activity" is in fact related to the task they did in the scanner (some voxels really do light up), that activity is **completely unrelated to the psychological test in any brain voxel**. Let's see how often we'd be fooled into thinking the brain activity and test scores are related, when we know they are not.

Question 1. Replicate a common data analysis workflow to identify brain areas whose activity correlates with a psychology test result.

1. Find out how correlated the Differential Activity in each voxel is with the subjects' psychology test scores. For every one of the N_{vox} voxels in turn, take the 12 subjects' differential activity value and the 12 subjects' psychology test scores T, and test if these are correlated. This gives you N_{vox} r values and p values.



2. Define the "Modulated" voxels to be all the voxels that are **positively correlated** with the **psychology test scores** $\boldsymbol{T}$ with the criteria r>0.1 and p<0.05. (Note that these may or may not be task-dependent voxels).
3. **Within** each subject, average the Differential Activity of the "Modulated" voxels defined in step 2 to get a single "Relevant Brain Network Activity" value for each subject.
4. Make a scatter plot of the Relevant Brain Network Activity of each subject versus their psychology test score T (1 symbol per subject, 12 subjects).
5. Plot the best fit line through the points, and label the plot with the R^2 value and p value. Did you find an effect? Did it look convincing on the graph? Were the R^2 value and p value convincing?
6. Optional: run that code 1000-10000 times, simulating a new random null data set each time, and determine what fraction of experiments produce a "significant" (p<0.01) result.

Question 2 Repeat the entire simulation above, but this time simulate that the outcome of the psychology test really *does* predict *how strongly* the test condition in the scanner modulates brain activity in the task-dependent brain voxels. To achieve this, instead of using μ_A for the task-dependent voxels when you simulate the differential activity, use $\mu_A(i)$=$f(T(i))$ where task response magnitude is some function of the subject's test score T. Contrive it so that the average value of $\mu_A(i)$ is the same as the fixed μ_A in the null simulation, and $\mu_A(i)$ is noticeably but only modestly correlated with $T(i)$.

Repeat the analysis workflow as above. Compare the plots you get (step 5), or the chance of getting a "significant" finding (from step 6).

What to think about: did anyone get a false positive (i.e. discover a brain network that was significantly correlated with the psychology test) when you simulated the null hypothesis (Question 1)? If so, how convincing was it – either in terms of R^2 or P value or scatter plot? Compare the scatter plots, R^2 and P values from Q1 and Q2. How could one tell a false positive from a real effect?

[Plot spoiler: everyone will get a false positive; and the true positives look pretty similar]

Question 3: What is the problem with this data analysis workflow?

If you think you know what's wrong with the analysis procedure described above, can you perform the analysis in Q1 in some better way, so that you're less likely to get a false positive, or so any false positives you get would be less convincing? If you apply this improved method in Q2, are you still able to detect a real effect? The data should be simulated the same way; only change the analysis procedures. Your test of the proposed improvement can be qualitative (i.e. test the new method in one run) or quantitative (repeat a lot of times to estimate FP rate and statistical power).